

HW7

Review Questions and Exercises

1. Size declarator defines an array, specifying the total number of elements; Subscript access a single element in the array.

3. When an array is passed to a function, only the array's starting address is passed, not the size.

The function requires an extra parameter to specify the number of elements to prevent out-of-bounds.

4. 2 ; 14 ; 8

7. In C/C++ an array name is a constant pointer that holds the array's starting address. The simple assignment attempts to copy the address, but an array's address cannot be reassigned. To copy the contents, must use loop to copy elements individually.

9. reference

15. size declarator

16. integer ; zero

18. $N-1$

31. 6 ; 5

52. T

60. F

82. Out of bounds and starting index is 0

should be `int i=0 ; i<=4 ; i++`

83. all elements should be int out of bounds, only 4 elements

Programming Challenges

10.

```
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     const int SIZE = 20;
7     char correctAnswers[SIZE] = {'A', 'D', 'B', 'B', 'C', 'B', 'A', 'B', 'C', 'D', 'A', 'C', 'D', 'B', 'D', 'C', 'C', 'A', 'D', 'B'};
8     char studentAnswers[SIZE];
9     int counter = 0;
10    char answer;
11
12    cout << "Enter the student's answers for the 20 questions (A, B, C, or D):" << endl;
13    for (int i = 0; i < SIZE; i++)
14    {
15        cout << "Question " << (i + 1) << ": ";
16        cin >> answer;
17        if (answer != 'A' && answer != 'B' && answer != 'C' && answer != 'D')
18        {
19            cout << "Invalid input. Please enter A, B, C, or D: ";
20            cin >> answer;
21        }
22        else
23        {
24            studentAnswers[i] = answer;
25        }
26    }
27
28    for (int i = 0; i < SIZE; i++)
29    {
30        if (studentAnswers[i] == correctAnswers[i])
31        {
32            counter++;
33        }
34    }
35
36    if (counter >= 15)
37    {
38        cout << "The student passed the exam." << endl;
39    }
40    else
41    {
42        cout << "The student failed the exam." << endl;
43    }
44
45    cout << "Total correct answers: " << counter << endl;
46 }
```

```
Enter the student's answers for the 20 questions (A, B, C, or D):
Question 1: A
Question 2: D
Question 3: B
Question 4: B
Question 5: C
Question 6: B
Question 7: A
Question 8: B
Question 9: C
Question 10: D
Question 11: A
Question 12: C
Question 13: D
Question 14: B
Question 15: D
Question 16: C
Question 17: C
Question 18: A
Question 19: D
Question 20: B
The student passed the exam.
Total correct answers: 20

-----
Process exited after 28.04 seconds with return value 0
请按任意键继续. . .
```